

In questions 1-3, determine if each of the following represents a definition, postulate, or theorem. Then, draw a diagram using the given information for each representation.

1. If two supplementary angles are congruent, then they are right angles.
2. A midpoint divides a segment into two congruent parts.
3. If a point, B , is drawn on a line segment AC , then AC is the sum of AB and BC .

Complete the table.

	Property	Equality	Congruence
4.		If $m\angle ABC = m\angle DEF$ and $m\angle DEF = 45^\circ$, then $m\angle ABC = 45^\circ$	If $\angle ABC \cong \angle DEF$ and $\angle DEF \cong \angle GHI$, then $\angle ABC \cong \angle GHI$
5.	Reflexive	$m\angle JKL = m\angle JKL$	
6.	Symmetric	If $m\angle XYZ = 81^\circ$, then _____	If $\angle XYZ \cong \angle FGH$, then $\angle FGH \cong \angle XYZ$

7. Write the conclusion for the statement.

If $m\angle JKL + m\angle SMW = m\angle XYZ + m\angle DNF$ and $m\angle XYZ = m\angle SMW$, then...

8. Name the property to justify the statement in question 7.

Complete the proof.

	Statement	Reason
9.	$m\angle MNP = m\angle DFG$	
10.		Definition of Congruence

Geometry

Unit 2

Lesson 2 Homework

Name_____

Date_____

Period_____

In questions 1-5, determine the type of reasoning for each statement.

1. In order to participate in school sports, students must have a minimum GPA of a 2.0. Jim has a GPA of 2.75, therefore he can participate in school sports.

2. Every Tuesday that Yolanda's family has visited a specific local restaurant, the kids' meals have been free. When Yolanda's family visits that restaurant next Tuesday, the kids' meals will be free.

3. A new movie director has grossed over 250 million for each of his first three movies. When his fourth movie is released, it will gross over 250 million.

4. Any given number multiplied by one equals itself. If 5 is multiplied by one, it will equal itself.

5. The air conditioner is set to keep a particular home at 72°F and runs off the home's electricity. During a power outage, the home will not stay at 72°F.

For each statement in 6-10, write a counterexample that disproves the statement.

6. Mary enters her home and her clothes are wet. Therefore, it is raining outside.

7. If a number is multiplied to another number, it will result in a larger number.

8. If a figure has four sides and all sides have the same length, it is a square.

9. If a number is prime, then it is an odd number.

10. Yvette lives in Jacksonville. Therefore, she lives in Florida.

Geometry
Unit 2
Lesson 3 Homework

Name _____
Date _____
Period _____

In the game of “Crazy Eights”, each player is dealt 7 cards. The remainder of the stack is placed face down on the table. The top card from the stack is flipped over and placed face up on the table. The first player must either match the suit (Diamonds, Clubs, Spades, or Hearts), the number (1,2,3,4,5,6,7,8,9,10) or the face card (Jack, Queen, King, Ace).

Using these rules, complete the table to end at a King of Clubs.

	Statement	Reason
1.	Ace of Spades	
2.		
3.		
4.		
5.		

	Statement	Reason
6.		
7.		
8.		
9.		
10.	King of Clubs	

Geometry

Unit 2

Lesson 4 Homework

Name_____

Date_____

Period_____

For questions 1-3, use the quote shown.

“If you build it, they will come” – Kevin Costner from the film, *Field of Dreams*

1. Write the converse of this statement.

2. Write the inverse of this statement.

3. Write the contrapositive of this statement.

For questions 4-6, use the quote shown.

“When you’re curious, you find lots of interesting things to do” – Walt Disney

4. Write the converse of this statement.

5. Write the inverse of this statement.

6. Write the contrapositive of this statement.

For questions 7-8, use the quote shown.

"If I had asked people what they wanted, they would have said faster horses". - Henry Ford

7. What is the hypothesis?

8. What is the conclusion?

Rewrite the following statements as conditional statements.

9. When the going gets tough, the tough gets going.

10. When you are not first, you are not last.

Geometry

Unit 2

Lesson 5 Homework

Name _____

Date _____

Period _____

Complete each table for the conditional statement.

If an angle is classified as a right angle, then it has a measure of 90° .			
	Relationship to Conditional	Statement	True or False
1.	Converse		
2.	Inverse		
3.	Contrapositive		
Write the conditional using "if and only if".			
4.			

If two angles have the same measure, then they are congruent			
	Relationship to Conditional	Statement	True or False
5.	Converse		
6.	Inverse		

7.	Contrapositive		
Write the conditional using “if and only if”.			
8.			

9. Write a conditional statement that is biconditional.

10. Write a conditional statement that is **not** biconditional.

Geometry
Unit 2
Lesson 6 Homework

Name _____
Date _____
Period _____

Determine if each statement is true. Rewrite each false statement using the word "some".

	Statement	True or False	Re-write
1.	All even numbers are composite.		
2.	All points that lie in the same plane are coplanar.		
3.	All linear pairs are supplementary.		

Write each statement as an if-then statement.

	Statement	If-Then
4.	All even numbers are composite.	
5.	All points that lie in a plane are coplanar.	

6.	All linear pairs are supplementary.	
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Write the negation of each statement.

	Statement	Negation
7.	$\triangle ABC$ is an equilateral triangle.	
8.	$\angle DEF$ and $\angle KLM$ are supplementary angles.	
9.	$\angle JKL$ and $\angle XYZ$ are vertical angles.	
10.	$m\angle DEF + m\angle KLM = 125^\circ$	

Geometry
Unit 2
Lesson 7 Homework

Name _____
Date _____
Period _____

Complete the proof.

Given: $AB = CD$

Prove: $AC = BD$



	Statement	Reason
1.	$AB = CD$	
2.		Reflexive Property of Equality
3.	$AB + BC = BC + CD$	
4.		Segment Addition Postulate
5.		Substitution Property of Equality

Complete the paragraph proof.

Given: $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are complementary.

Prove: $\angle 1 \cong \angle 3$

It is 6. _____ that $\angle 1$ and $\angle 2$ are complementary and $\angle 2$ and $\angle 3$ are

complementary. By the definition of 7. _____,

$m\angle 1 + m\angle 2 = 90^\circ$ and $m\angle 2 + m\angle 3 = 90^\circ$. By the 8. _____ property,

$m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$.

Then, $m\angle 1 =$ 9. _____ by the subtraction property. Therefore, $\angle 1 \cong \angle 3$ by the

10. _____.

Complete the two-column proof.

Given: $\overline{AB} \cong \overline{CD}$

Prove: $\overline{AC} \cong \overline{BD}$



	Statement	Reason
1.	$\overline{AB} \cong \overline{CD}$	Given
2.		Reflexive Property
3.		
4.	$AB + BC = AC$ $CD + BC = BD$	
5.	$CD + BC = AC$	Substitution Property
6.		Substitution Property
7.	$\overline{AC} \cong \overline{BD}$	

A partially completed two-column proof is shown. Match each statement on the left with the correct reason to complete the two-column proof by using reasons provided in the bank. Some reasons may be used more than once.

Given: W is the midpoint of $\overline{VX}$, Z is the midpoint of $\overline{RP}$, and $\overline{VX} \cong \overline{RP}$

Prove: $\overline{WX} \cong \overline{ZP}$



Reasons			
Given	Combine like terms	Definition of congruence	Definition of midpoint
Division property of equality		Segment addition postulate	Substitution property of equality

	Statement	Reason
1.	W is the midpoint of $\overline{VX}$, Z is the midpoint of $\overline{RP}$, and $\overline{VX} \cong \overline{RP}$	
2.	$\overline{VW} \cong \overline{WX}$ $\overline{RZ} \cong \overline{ZP}$	
3.	$VX = RP$	
4.	$VW + WX = VX$ $RZ + ZP = RP$	
5.	$VW = WX$ $RZ = ZP$	
6.	$VW + WX = RZ + ZP$	
7.	$WX + WX = ZP + ZP$	
8.	$2WX = 2ZP$	
9.	$WX = ZP$	
10.	$\overline{WX} \cong \overline{ZP}$	